

SWITCH AND ROUTER CONFIGURATION

BASIC THEORY

1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.



```
Press RETURN to get started!

Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan10
^
% Invalid input detected at '^' marker.

Switch(config)#vlan10
^
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name sales
VLAN #10 and #20 have an identical name: sales
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Switch#show vlan brife
^
% Invalid input detected at '^' marker.

Switch#show vlan brief

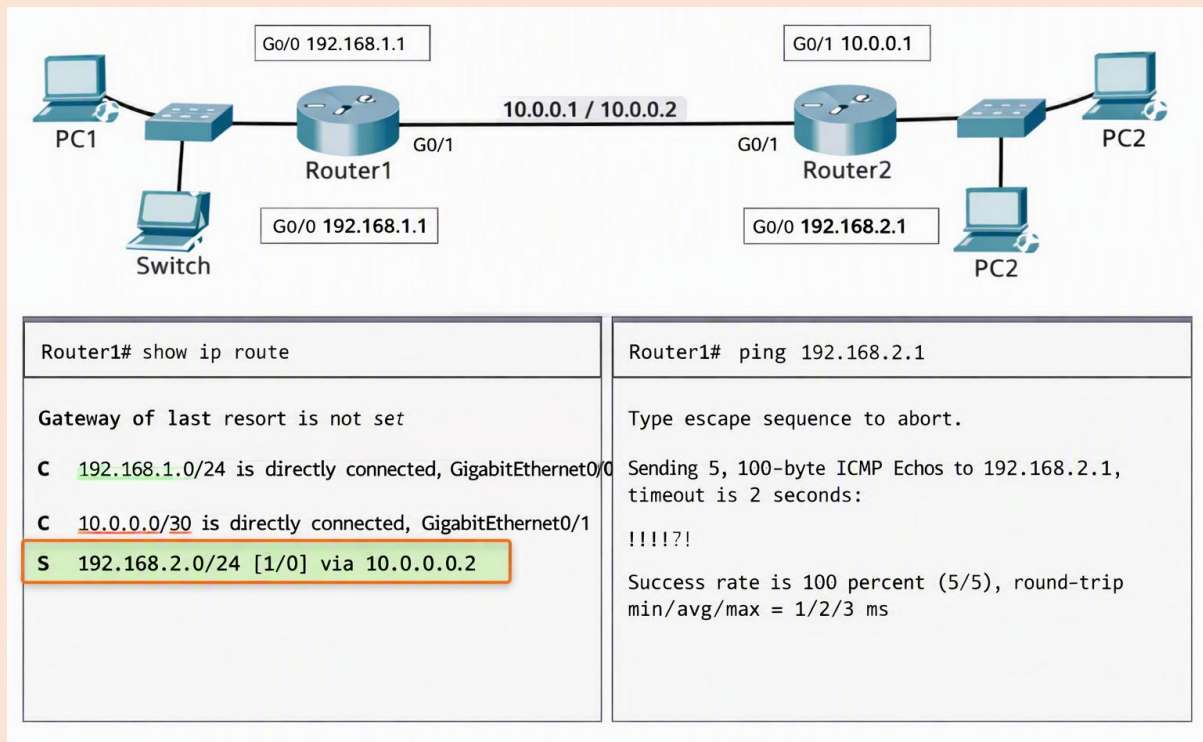
VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   sales                  active
20   VLAN0020               active
1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default       active
1005 trnet-default         active
Switch#
```

2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.
Hint: Use the interface range command to select multiple ports efficiently.



```
VLAN Name                Status    Ports
-----
1    default                active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                           Gig0/2
10   sales                  active    Fa0/1, Fa0/2, Fa0/3
20   Support                active    Fa0/4, Fa0/5
1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default       active
1005 trnet-default         active
Switch#
```

3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.



4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. **Hint:** Think about router-on-a-stick or Layer 3 switch concepts.



Scenario:

In a network, two different VLANs are created:

- VLAN 10: Sales Department
- VLAN 20: Support Department

Example IP Address:

Device	VLAN	IP Address
Sales PC	VLAN 10	192.168.10.10
Support PC	VLAN 20	192.168.20.10

When the Sales PC tries to ping the Support PC using the command:

ping 192.168.20.10

the ping request fails.

Reason for Ping Failure:

The ping fails because VLANs are separate broadcast domains. A Layer 2 switch cannot forward traffic directly between different VLANs.

VLAN 10 and VLAN 20 have different network addresses, so communication between them requires a Layer 3 device (Router or Layer 3 Switch) to perform routing.

Without inter-VLAN routing configuration, devices in different VLANs cannot communicate with each other.

Solution: Enable Inter-VLAN Communication

Inter-VLAN communication can be achieved using:

1. Router-on-a-Stick Method:

A router is connected to the switch using a trunk link. The router creates separate sub-interfaces for each VLAN.

2. Layer 3 Switch Method:

A Layer 3 switch can perform routing internally by creating Switch Virtual Interfaces (SVIs).

After enabling routing, devices in Sales VLAN and Support VLAN can communicate.